FRONTEND PERFORMANCE OPTIMIZATION REPORT

Week 4 Challenge · Evergreen Performance / Collection page

Website under review	https://puneet-17-ai.github.io/Evergreen-Performance/#collection
Prepared	28 October 2026
Report status	Evidence template — complete after running actual audits

Important: no performance values have been invented. Complete the report using genuine Lighthouse measurements from the original and revised websites. The included source is a standalone demonstration project, not a verified patch to the live repository.

1. Objective and methodology

Measure front-end performance with Google PageSpeed Insights / Lighthouse, identify evidenced bottlenecks, apply suitable optimizations, and compare the baseline and revised versions. Run mobile and desktop tests on the same URLs and save the reports or screenshots. Lighthouse results may vary between runs; report actual values and comparable conditions.

2. Baseline measurements — pending

Metric	Mobile baseline	Desktop baseline
Performance score	PENDING	PENDING
Accessibility score	PENDING	PENDING
Best Practices score	PENDING	PENDING
SEO score	PENDING	PENDING
FCP	PENDING	PENDING
LCP	PENDING	PENDING
Speed Index	PENDING	PENDING
TBT	PENDING	PENDING
CLS	PENDING	PENDING

3. Bottlenecks to verify

Check the actual Lighthouse diagnostics for render-blocking resources, oversized images, unused JavaScript, main-thread work, layout shifts, third-party scripts, font requests, and caching. Only list a bottleneck as confirmed if supported by the baseline report or source inspection.

Confirmed findings: PENDING — add Lighthouse audit names and evidence.

4. Optimizations in the included demo

The demo uses semantic HTML, a skip link, a small dependency-free JavaScript file loaded with defer, a system font stack, CSS-generated decorative artwork, responsive layouts, and reduced-motion support. Merge applicable changes into the original source before describing them as changes to the submitted website.

5. Post-optimization measurements — pending

Metric	Mobile after	Desktop after
Performance score	PENDING	PENDING
Accessibility score	PENDING	PENDING
Best Practices score	PENDING	PENDING
SEO score	PENDING	PENDING
FCP	PENDING	PENDING
LCP	PENDING	PENDING
Speed Index	PENDING	PENDING
TBT	PENDING	PENDING
CLS	PENDING	PENDING

6. Comparison and conclusion

Fill in baseline, after, and change values once both audits are complete. For scores, change = after − baseline. For time metrics, improvement = baseline time − after time. Do not claim percentage improvements without real values. This report is not submission-complete until it contains actual audit evidence.

7. Evidence checklist

■ Baseline mobile report/screenshot

■ Baseline desktop report/screenshot

■ Optimized mobile report/screenshot

■ Optimized desktop report/screenshot

■ Updated report and actual revised source in ZIP

References

Google PageSpeed Insights overview: developers.google.com/speed/docs/insights/v5/about
Lighthouse documentation: developer.chrome.com/docs/lighthouse/
Audit tool: pagespeed.web.dev/